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Smart Bike Lanes: The Future of Micromobility at Georgia Tech

The topic of micromobility at Georgia Tech is a divisive subject. Many students applaud the expansion of micromobility infrastructure. However, critics have pointed out the downsides of proliferating micromobility infrastructure. Many serious collisions have occurred between pedestrians, scooters, and bikes. Furthermore, while micromobility devices may be better for the environment than car travel, many scooters and bikes demanded charging infrastructure and separate pavement space from cars, leading to environmental impacts on the Georgia Tech community.

The growth of micromobility at Georgia Tech reinforces the university's leadership in urban innovation, but further improvements are necessary to ensure new infrastructure equitably and safely accommodates all students. In the 2012 GT Campus Bicycle master plan, it is noted that the number of commuters already exceeds current bike rack capacity (Georgia Institute of Technology 2015). Despite the ample warning, this problem has not only persisted, but worsened. Georgia Tech needs flexible micromobility infrastructure that acts proactively to users' needs rather than reacting to decades-long issues.

In 2014, in the Netherlands, the first solar-powered bike path was built, SolaRoad. This 70 meter stretch of road had integrated solar panels into its surface, which generated 98,000 kWh of energy in its first year alone. Then, in 2023, two new bike paths, also inspired by SolaRoad, were inaugurated in the Netherlands as well. These new bike paths covered 1,000m^2 and used the same technology, generating 140 MWh, or 140,000 kWh annually (enough to power 57-64 avg.

Dutch homes) (Santa Fixie 2026). Many European companies and government agencies have been inspired to begin implementing this technology, which further proves its impressive efficiency and scalability.

Having seen the success of solar bike path projects in the Netherlands, we were inspired to install solar panels and movement sensors across Georgia Tech, called Solar Bike Lanes. The goal of doing this is to not only serve the needs of the Georgia Tech community, but also make something that is scalable to other universities and cities across America.

The solar panels would function similarly to the solar panels bike paths in the Netherlands. A key difference between Atlanta and the Netherlands is that Atlanta receives on average one hour of sunlight more than the Netherlands per month, so the solar panels would be more effective at Georgia Tech (ClimeChart 2026). The additional energy generated by the solar panels can be used to illuminate the path followed by cyclists in times of low light. This illumination with LED lights will be a wave of medium intensity to ensure it doesn't blind cyclists, while still serving as a guide and road safety aid. The energy that is not used bike path lighting will be fed into scooter and e-bike charging stations. This would work as a microgrid of direct current since solar panels generate energy in a direct current. Bike and scooter batteries are also charged in DC, therefore we would avoid energy losses that happen when converting from AC (alternating current) to DC (direct current).

In addition to the presence of solar panel bike lanes, our proposal will also include installing bike detection wires beneath bike paths as well. This will allow Parking and Transportation Services (PTS) to analyze data about how many vehicles are using the bike lanes and make estimates about future bike lane demands (FHWA 2007) Now, Georgia Tech can finally design its infrastructure proactively for micromobility users. Furthermore, the collected data will be interpreted in an app that tells bike and scooter users where areas of high and low traffic are. Micromobility users can then plan their trips around traffic patterns, which will decrease nonessential traffic on paths during peak travel hours. PTS can also use this data to know where and when station officers to manage traffic and decrease the likelihood of collisions.

By implementing our proposal of solar powered bike/scooter lanes and live tracking data of lane users with underground wires, Georgia Tech can accurately access and evaluate where micromobility infrastructure will be most effective. Active participation by micromobility users will create an accurate and effective data collection system that will benefit the entire community. They will feel represented and heard by Georgia Tech administration due to seeing the real-time effects of our work through adaptive infrastructure resulting from data collection. Smart Bike Lanes will propel Georgia Tech into a new echelon of innovation and leadership in micromobility infrastructure amongst other universities and the nation.

Bibliography

Georgia Institute of Technology. 2015. "Georgia Tech Campus Bicycle Master Plan." *Georgia Institute of Technology Facilities.* Accessed February 25, 2026. https://facilities.gatech.edu/sites/default/files/2023-10/GT%20Campus%20Bicycle%20Master%20Plan_lowres.pdf

This source was a former master plan that outlined goals for future cycling growth at Georgia Tech while analyzing data pertaining to high traffic corridors around Georgia Tech. It contains many figures related to predicted growth, which were heavily used in this project proposal paper.

ClimeChart. 2026. "Climate Comparison: Amsterdam, Netherlands vs. Atlanta, United States of America." *ClimeChart.* Accessed February 25, 2026. https://www.climechart.com/en/climate-compare/amsterdam/netherlands/atlanta/united-states-of-america

This source provided comparison data between the city of Amsterdam (capital of the Netherlands), and Atlanta. This data was used to determine how much extra sunlight would be absorbed by the solar panels in Atlanta in comparison to the Netherlands, where the solar bike path project was first launched.

Santa Fixie. 2026. "Solar Bike Lane Impact & Sustainable Mobility." *SantaFixie.* Accessed February 25, 2026. https://www.santafixie.co.uk/bloguk/solar-bike-lane-impact-sustainable-mobility/

This source presents a Dutch solar bike lane project, which was the first of its kind, known as SolaRoad. It provided information regarding the amount of energy generated by it over the course of 2014. It later mentions similar projects that were inspired by SolaRoad and their success at a fairly larger scale.

Federal Highway Administration (FHWA). 2007. "Inductive Loop Detectors." *Federal Highway Administration.* Accessed February 25, 2026. https://www.fhwa.dot.gov/policyinformation/pubs/vdstits2007/04.cfm#:~:text=INDUCTIVE%20LOOP%20DETECTORS

This source provided the geometry and overall description of deploying sensory wires on bike paths to track traffic numbers amongst micromobility users. We used this source to understand the challenges and benefits of tracking micromobility users and how we can use this data to benefit the larger Georgia Tech community as a whole.